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DIAGNOSTIC METHODS: OBSERVATIONAL, CROSS-SECTIONAL, STUDY

Evaluation of integral electromyographic values and median power frequency values in women with myogenous temporomandibular disorder and asymptomatic controls

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ABSTRACT

Background: Surface electromyography (EMG) has been used as a reliable tool for the evaluation of electrical muscle activity.
Objective: The purpose of this study was to evaluate the EMG indices of the masticatory muscles (masseter, anterior temporalis and suprahyoid) in women with temporomandibular disorder (TMD) and asymptomatic controls in the time domain, by the integrated EMG signal (IEMG) and in the frequency domain, using the median power frequency (MPF).
Methods: An observational, cross-sectional study was conducted involving 30 asymptomatic women as the control (mean age: 25.85 ± 2.57 years) and 74 women with myogenous TMD (mean age: 26.54 ± 2.45 years) diagnosed using the Research Diagnostic Criteria for Temporomandibular Disorders (RDC/TMD). Three EMG readings were taken during maximum voluntary clenching (MVC) of the molars on parafilm for five seconds with a three-minute rest interval between readings. The mixed-model analysis of variance test followed by the Bonferroni correction or the Student-t test was used for the analyses, with the level of significance set to 5% ($p < 0.05$).
Results: IEMG values were significantly higher in the masseter muscles than the anterior temporalis muscles in the control group ($p < 0.01$). IEMG values were significantly higher in the masseter muscles of the control group than the group with myogenous TMD ($p < 0.05$). MPF values of the suprahyoid muscles were significantly higher in the myogenous TMD group than the control group.
Conclusions: These significant findings show that women with myogenous TMD have reduction of electrical activity of their masseter muscles and increased firing rate of the motor units of the suprahyoid muscles. These findings may help the treatment of myogenous TMD in women.
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1. Introduction

Temporomandibular disorder (TMD) is a set of clinical alterations involving the masticatory muscles and temporomandibular joint (TMJ) (Leeuw and Klasser, 2013). Signs and symptoms of this disorder include limited jaw movement, sensitivity of the masticatory muscles and TMJ, joint sounds (clicks and crepitus), ringing in the ears, headache and upper neck pain (Ciancaglini and Radaelli, 2001; Kordass et al., 2012). Moreover, previous studies have demonstrated that TMD is more prevalent in the female gender (Slade et al., 2011; Slade, 2014).

TMD leads to important changes in the electrical activity of the masticatory muscles due to the disorder itself or compensatory mechanisms associated with the symptoms (Santana-Mora et al., 2009), since the main problems in affected individuals are of a muscular origin (Santana-Mora et al., 2009; Douglas et al., 2010). Thus, surface electromyography (EMG) is an adequate tool for the analysis of neuromuscular function in such cases. When used following specific guidelines and in conjunction with an analysis of the clinical history and physical exam of the patient, EMG provides objective, valid, reproducible data on the functional condition of the masticatory muscles in individuals with TMD (Hugger et al., 2012). Moreover, this is a noninvasive, low-cost technique that

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can be used in the clinical setting without the need to send the patient to a specialized diagnostic center (Sforza et al., 2011; Lodetti et al., 2012).

It is known that the electrical activity of a given muscle (EMG) can be processed according to specific calculations using the root mean square (RMS) and integrated EMG signal (IEMG) in the time domain, as the median power frequency (MPF) of power spectral density in the frequency domain (De Luca, 1997). In this context, studies have documented normalized and non-normalized RMS values of the masseter and anterior temporalis muscles with the jaw at rest and/or during maximum voluntary clenching (MVC) of the molars either in intercuspal position (Santana-Mora et al., 2009; Liu et al., 1999) or on cotton rolls (Tartaglia et al., 2011; De Felício et al., 2012) or on parafilm (Berni et al., 2015; Lauriti et al., 2013) in individuals with TMD and asymptomatic individuals. However, only Santana-Mora et al. (2009) and Liu et al. (1999) have reported IEMG values of the masseter and anterior temporalis muscles of this population during MVC with the molars in intercuspal position. Thus, there are no studies in the literature on normalized or non-normalized IEMG values during MVC on cotton rolls or parafilm.

In addition, studies have demonstrated that the MPF offers a lower probability of being influenced by subjective aspects of the individual, such as cooperation, mood and psychological factors, thereby enabling a better interpretation of the specific electrical behavior of masseter and anterior temporalis muscles (Tartaglia et al., 2011; De Felício et al., 2012; Lauriti et al., 2013). It is therefore, important to evaluate IEMG values together with MPF values for the characterization of individuals affected with myogenous TMD and asymptomatic individuals.

Another important point to consider regards the lack of information in the recent literature on the IEMG and MPF of the suprahyoid muscles during MVC in individuals with TMD, despite the fact that these muscles are considered important to the mechanics of the TMJ as agonists of the mouth-opening movement (Standring, 2011). Thus, the present investigation was conducted to test the hypothesis that significant differences occur in the EMG signals of the masticatory muscles (masseter, anterior temporalis and suprahyoid muscles) between individuals with myogenous TMD and asymptomatic individuals during MVC.

The aim of the present study was to evaluate IEMG (normalized and non-normalized) and MPF values of the masticatory muscles in women with myogenous TMD and asymptomatic controls during MVC on parafilm.

2. Methods

2.1. Sample

An observational, cross-sectional study was conducted with women: 74 with a diagnosis of myogenous TMD (mean age: 26.54 ± 2.45 years) allocated to the TMD group and 30 asymptomatic women allocated to the control group (mean age: 25.85 ± 2.57 years). The inclusion criteria for the group with TMD were age 18–40 years, diagnosis of myogenous TMD [myofascial pain (Ia) or myofascial pain with limited mouth opening (Ib) according to the Research Diagnostic Criteria for Temporomandibular Disorders (RDC/TDM)], bilateral TMJ dysfunction, pain and/or fatigue in the masticatory muscles during functional activities for more than six months (Ohrbach et al., 2011) and body mass index (BMI) between 18 and 25 kg/m^2, because the amount of adipose tissue between the surface of the muscle and electrode can affect the capture of the EMG signal (De Luca, 1997). Moreover, women with simultaneous diagnoses of disk displacement or arthralgia (IIa, IIb, IIc and IIIa) were permitted in TMD group, according with the RDC/TMD. The inclusion criteria for the control group were age 18–40 years, the absence of TMD, according with the RDC/TMD and BMI between 18 and 25 kg/m^2.

The following were the exclusion criteria for both groups: tooth loss, body mass index (BMI) greater than 25 kg/m^2, systemic disease (such as arthritis, arthrosis or diabetes), history of trauma to the face or TMJ, history of sub-luxation or luxation of the TMJ, current orthodontic treatment or current medical treatment involving anti-inflammatory agent, analgesic or muscle relaxant and volunteers with a diagnosis of arthrogenous TMD (IIIb and IIIc) according with the RDC/TMD.

Between 2009 and 2013, 102 women with TMD and 50 asymptomatic women (without TMD) were recruited from the university community of the city of Piracicaba (SP, Brazil) at the Methodist University of Piracicaba, through personal invitation, posters, and Internet advertisements. Of the women with TMD, 28 were excluded: twelve had a diagnosis of IIIb or IIIc on the RDC/TMD, eight were in orthodontic treatment, and eight had a body mass index higher than the established cutoff point. Of the asymptomatic women, 20 were excluded, since ten were in orthodontic treatment and ten had a body mass index higher than the established cutoff point.

The procedures received approval from the Human Research Ethics Committee of the Methodist University of Piracicaba (SP, Brazil), under protocol number 25/15. All volunteers signed a statement of informed consent.

2.2. Evaluations

The RDC/TMD was used to screen the volunteers of the groups. Pain in the TMJ was evaluated using the visual analog scale, which is a tool that facilitates the measurement of pain intensity and consists of a straight line 10 cm in length on which the left extremity represents the absence of pain and the right extremity represents the worst possible pain (Ferreira-Valente et al., 2011; Packer et al., 2014). The volunteers were instructed to mark a perpendicular line between the two extremes that represented the pain intensity at rest that they were feeling at the moment of recruitment.

The BIO-EMG 1000 electromyograph (Lynx®, Tecnologia Eletrônica Ltda., São Paulo, SP, Brazil) was used for the surface EMG procedures, with impedance of 109 Ω, an analog-digital converter with 16 bits of resolution, a sampling frequency of 2000 Hz per channel and a range of ±2 V. The specific software program for this equipment was used for the acquisition and storage of the digitized signals.

Five simple differential electrodes were used to capture the action potential of the masseter (right and left), anterior temporalis (right and left) and suprahyoid muscles. The electrodes consisted of two pure silver bars measuring 10 × 2 mm positioned parallel, with a fixed inter-electrode distance of 10 mm, 1000-fold gain, CMRR greater than 130 dB, input impedance of 10 GΩ and noise/signal ratio less than 3 μV RMS. After clearing the site of hair follicles and cleaning with a cotton ball moistened with 70% alcohol, the electrodes were positioned over the belly of the muscles with the silver bars perpendicular to the muscle fibers following the method described by Cram et al. (1998). A reference electrode composed of a metal plate measuring 30 × 40 mm was positioned over the manubrium of the sternum. During the readings, the volunteers remained seated in a chair and performed a maximum voluntary teeth clenching (MVC) on parafilm, in which the subject was invited to clench the molar teeth as hard as possible and to maintain the same level of contraction for 5 s. Three five-second readings were made with a three-minute rest interval between readings (Berni et al., 2015).

2.3. Processing of EMG signal

The signals were analyzed offline with the aid of the Matlab® 8.3 program (The MathWorks Inc., Natick, Massachusetts, USA). A zero phase delay 4th order Butterworth digital filter was used, with a high band pass of 10 Hz and low band pass of 400 Hz. The first and fifth seconds of each EMG reading were discarded to avoid any interferences that may have occurred at the beginning and end of the signal capture procedure. Thus, the middle three seconds of each signal were considered in the analysis.

EMG analysis was performed 1) in the frequency domain for the determination of the MPF (Hz) of the power spectral density of the signal, which was evaluated using the fast Fourier transform algorithm with 50% overlap (Hamming window processing), and 2) in the time domain for the determination of IEMG values using a 150-ms RMS moving window with subsequent integration. The data in time domain was presented by IEMG non-normalized values (μV.s) and, also, by normalized IEMG values (IEMGn) expressed as percent of the maximal potential recorded from the three MVC readings on parafilm (unit: μV.s/maximum μV.s x 100). The means of the MPF, IEMG and IEMGn of each three readings were calculated for the purposes of analysis.

2.4. Statistical analysis

The data were expressed as mean and standard deviation values. Mean differences between groups were expressed with respective 95% confidence intervals. As the quantitative data (IEMG, IEMGn and MPF of the masticatory muscles) demonstrated normal distribution and homoscedasticity, a mixed-model analysis of variance test (ANOVA) was used, with between subjects factors: group (TMD and control) and within subjects factors: muscle (masseter and anterior temporalis) and side (right and left). The main effects (group, muscle and side) and interactions (group x muscle, group x side, muscle x side and group x muscle x side) were evaluated, followed by the Bonferroni correction for significant interactions. The Student's t-test was used for the analysis of quantitative data of the suprahyoid muscles, since these muscles were only evaluated at a single point and not on the right and left sides. Moreover, the suprahyoids are considered antagonist muscles during MVC, making their comparison to elevator muscles unnecessary. Intra-examiner reliability was evaluated using the intraclass correlation coefficient ($ICC_{3,1}$), which was interpreted as follows: low reliability <0.40; good reliability ≥0.40 and ≤0.75; and excellent reliability >0.75 (Munro, 2001). The standard error of measurement (SEM) was calculated as follows: $SEM = standard\ deviation\ x\ \sqrt{(1 - ICC)}$. The SPSS® program, version 13.0 (Chicago, IL, USA), was used for all statistical analyses, with the level of significance set to 5%.

3. Results

Seventy-four volunteers were diagnosed with TMD: myofascial pain (Ia) or myofascial pain with limited mouth opening (Ib) and simultaneous diagnoses of disk displacement and arthralgia. TMD was absent in 30 volunteers, who then comprised the control group (see Table 1).

The groups were similar with regard to age and BMI. The TMD group generally exhibited mild pain intensity in the TMJ (see Table 2).

The Intra-session reliability of the IEMG data, assessed via intraclass correlation coefficient (ICC) and standard error of measurement (SEM) during MVC on parafilm, were ICC = 0.95 and SEM = 297.90 for right masseter muscle; ICC = 0.94 and SEM = 295.26 for left masseter muscle; ICC = 0.95 and SEM = 224.05 for right anterior temporalis muscle; ICC = 0.95 and SEM = 196.12 for left anterior temporalis muscle; and ICC = 0.77 and SEM = 48.49 for suprahyoid muscles. The Intra-session reliability of the MPF data were ICC = 0.88 and SEM = 37.66 for right masseter muscle; ICC = 0.92 and SEM = 27.27 for left masseter muscle; ICC = 0.94 and SEM = 27.08 for right anterior temporalis muscle; ICC = 0.85 and SEM = 35.98 for left anterior temporalis muscle; and ICC = 0.69 and SEM = 45.40 for suprahyoid muscles. Therefore, a good to excellent reproducibility was demonstrated for the IEMG and FM-EMG for the masticatory muscles during MVC on parafilm.

Table 1
Diagnosis of TMD according to the research diagnostic criteria for temporomandibular disorders.

Diagnosis	n
Without TMD – Control Group	30
Ia – TMD Group	53
Ib – TMD Group	21
Simultaneous diagnosis for the TMD group	
IIa (right/left TMJ)	22 (15/16)
IIb (right/left TMJ)	0 (0/0)
IIc (right/left TMJ)	5 (4/4)
IIIa (right/left TMJ)	38 (31/35)

TMD: Temporomandibular disorder; TMJ: Temporomandibular joint; Ia: Myofascial pain; Ib: Myofascial pain with limited mouth opening; IIa: Disc displacement with reduction; IIb: Disc displacement without reduction; IIc: Disc displacement without reduction and without limited mouth opening; IIIa: Arthralgia.

The descriptive statistics is presented in Table 3, IEMG (μV.s) and IEMGn (%) values, and Table 4, MPF values of the masticatory muscles during MVC on parafilm.

In general, IEMGn values were similar between jaw elevator (masseter and anterior temporalis) e and jaw depressors muscles (suprahyoid) considering the intragroup e intergroup analysis. On the other hand, IEMG values were higher in the masseter than in anterior temporalis muscles in the control group but not in the TMD group. The right side presented higher IEMG values in all jaw elevator muscles (independent of the group). The IEMG values of the suprahyoid muscles were similar between the groups and lower when compared to the jaw elevator muscles (see Table 3).

Table 4 shows lower MPF values in the masseter muscles than in anterior temporalis muscles (independent of the group or side) and even lower values in the suprahyoid muscles. Regarding to the MPF of the suprahyoid muscles, a lower value was observed in the control group than in TMD group.

Table 5 displays the inferential statistics on the EMG parameters presented in Tables 3 and 4, using mixed-model analysis of variance test for the jaw elevator muscles and the Student's t-test for the jaw depressors muscles. With regard to IEMG, a significantly greater electrical activity was found in the masseter muscles than the anterior temporalis muscles (without considering groups or sides) and significantly greater electrical activity was found on the right side than the left side (without considering groups or muscles). The MPF was significantly higher in the anterior temporalis muscles than the masseter muscles (without considering groups or sides). A significant group x muscle interaction was found for the IEMG values; after the Bonferroni correction, a significantly greater electrical activity in the masseter muscle was found in the control group when compared to the TMD group ($p < 0.05$), whereas no significant differences between groups were found with regard to the anterior temporalis muscles ($p > 0.05$). In addition, the electrical activity was significantly greater in the masseter muscles when compared to the anterior temporalis muscles ($p < 0.01$) in the control group after the Bonferroni correction, whereas no significant differences between muscles were found in the TMD group

Table 2
Anthropometric data and results of visual analog scale.

	TMD Group Mean ± Standard Deviation	Control Group Mean ± Standard Deviation	*p value*
AGE (years)	26.54 ± 2.45	25.85 ± 2.57	0.15
BMI (Kg/m^2)	23.56 ± 1.82	22.97 ± 1.98	0.1
VAS (cm)	3.2 ± 1.7	0 ± 0	Na

Student t-test.
Na: not applicable.
BMI: Body Mass Index.
VAS: Visual analog scale.

Table 3
Descriptive statistics of the IEMG (non-normalized) and IEMGn (normalized) values during MVC on parafilm in both groups.

GROUPS	Mean ± Standard Deviation	Differences in intergroup means (95% CI)
IEMGn – left anterior temporalis muscle (%)		
CONTROL	90.75 ± 5.23	−0.50 (−3.46–2.48)
TMD	91.25 ± 6.88	
IEMGn – left masseter muscle (%)		
CONTROL	89.95 ± 5.84	−0.48 (−3.47– 2.50)
TMD	90.43 ± 8.18	
IEMGn – right anterior temporalis muscle (%)		
CONTROL	91.34 ± 6.33	1.18 (−1.81 – 4.16)
TMD	90.16 ± 6.92	
IEMGn – right masseter muscle (%)		
CONTROL	90.12 ± 8.39	−0.21 (−3.19 – 2.77)
TMD	90.34 ± 6.68	
IEMGn – suprahyoid muscles (%)		
CONTROL	84.90 ± 10.83	−1.75 (−6.34 – 2.83)
TMD	86.66 ± 10.63	
IEMG – left anterior temporalis muscle (μV.s)		
CONTROL	411.11 ± 201.75	−38.76 (−197.66 – 120.14)
TMD	449.87 ± 338.36	
IEMG – left masseter muscle (μV.s)		
CONTROL	616.68 ± 360.00	85.09 (−73.81 – 243.99)
TMD	531.59 ± 404.16	
IEMG – right anterior temporalis muscle (μV.s)		
CONTROL	431.40 ± 228.90	−44.28 (−203,18 – 114.61)
TMD	475.69 ± 364.59	
IEMG – right masseter muscle (μV.s)		
CONTROL	764.07 ± 519.60	231.81 (72,91 – 390.72)
TMD	532.25 ± 416.00	
IEMG – suprahyoid muscles (μV.s)		
CONTROL	51.60 ± 19.65	−3.25 (−17.81 – 11.30)
TMD	54.86 ± 38.12	

95% CI: 95% confidence interval. The inferential statistics of the data in this table are presented in Table 5.

Table 4
Descriptive statistics of the Median power frequency (MPF) values during MVC on parafilm in both groups.

GROUPS	Mean ± Standard Deviation	Differences in intergroup means (95% CI)
MPF – left anterior temporalis muscle (Hz)		
CONTROL	162.02 ± 31.77	0.71 (−13.63 – 15.06)
TMD	161.30 ± 31.15	
MPF – left masseter muscle (Hz)		
CONTROL	137.17 ± 36.42	−4.47 (−18.81 – 9.87)
TMD	141.64 ± 28.27	
MPF – right anterior temporalis muscle (Hz)		
CONTROL	162.23 ± 36.81	7.18 (−7.17 – 21.52)
TMD	155.06 ± 36.67	
MPF – right masseter muscle (Hz)		
CONTROL	135.43 ± 28.71	−1,77 (−16,12 – 12.57)
TMD	137.21 ± 38.02	
MPF – suprahyoid muscles (Hz)		
CONTROL	85.50 ± 33.72	−14.08 (−27.20 – −0.96)
TMD	99.59 ± 29.20	

95% CI: 95% confidence interval. The inferential statistics of the data in this table are presented in Table 5.

($p > 0.05$). Regarding the suprahyoid muscles, a significant difference between groups was found for the MPF, with higher values in the TMD group.

4. Discussion

4.1. EMG analysis in time domain

No significant differences between groups were found regarding the normalized IEMG values for the masseter, anterior temporalis and suprahyoid muscles. Likewise, Lauriti et al. (2014) evaluated young adults with mild to severe TMD and asymptomatic individuals during MVC on parafilm and also found no significant differences between groups for the masseter and anterior temporalis muscles (the suprahyoids were not evaluated). However, the authors analyzed RMS normalized values rather than IEMG values.

With regard to non-normalized IEMG values, the masseter muscles generally demonstrated significantly greater activity than the anterior temporalis muscles. Moreover, this difference between muscles only occurred in the control group, suggesting that women with myogenous TMD exhibit a reduction in the electrical activity of the masseter muscles, with activity equaling that of the anterior temporalis muscles. This finding was confirmed in the inter-group analysis, as IEMG values for the masseter muscles were significantly greater in the control group than the TMD group.

Santana-Mora et al. (2009) and Liu et al. (1999) found significant differences between individuals with TMD and asymptomatic controls regarding non-normalized IEMG values for the masseter and anterior temporalis muscles, with lower values in the TMD group during MVC in intercuspal position. The same occurred in the present study with regard to the masseter muscles, but not the anterior temporalis muscles. This divergence may be due to the fact that the EMG signal was evaluated during MVC on parafilm in present investigation rather than in intercuspal position. Thus, the parafilm may have altered the mechanics of the TMJ, causing other motor unit recruitment patterns in the anterior temporalis muscles (Forrester et al., 2010). It should also be stressed that the IEMG values reported by Liu et al. were much higher than those in the present investigation for all muscles analyzed, also indicating different motor unit recruitment patterns in jaw elevator muscles when the bite was performed in intercuspal position.

Tartaglia et al. (2011) evaluated individuals with chronic arthrogenous TMD and asymptomatic controls and found significantly higher normalized RMS values for the anterior temporalis muscles during MVC. The differences found in comparison to the present investigation, in which individuals with myogenous TMD were evaluated, may have been due to the fact that the authors evaluated individuals with joint-related TMD, who exhibit greater dysfunction and pain intensity in the TMJ. Moreover, the authors evaluated MVC on cotton rolls and in intercuspal position rather

Table 5
Comparison of EMG values (ANOVA and Student's t-test) for the masticatory muscles considering the descriptive statistics of the data presented in Tables 3 and 4.

	IEMG	IEMG Normalized	MPF
ANOVA - (Group – main effect)	F = 0.80; $p = 0.37$	F = 0.00; $p = 0.99$	F = 0.31; $p = 0.57$
ANOVA - (Muscle – main effect)	F = 19.96; $p < 0.01$*	F = 1.38; $p = 0.24$	F = 50.86; $p < 0.01$*
ANOVA - (Side – main effect)	F = 5.21; $p < 0.05$*	F = 0.04; $p = 0.83$	F = 0.95; $p = 0.33$
ANOVA - (Group x Muscle)	F = 6.97; $p < 0.01$*	F = 0.36; $p = 0.54$	F = 0.01; $p = 0.88$
ANOVA - (Group x Side)	F = 2.57; $p = 0.10$	F = 0.90; $p = 0.34$	F = 0.50; $p = 0.48$
ANOVA - (Muscle x Side)	F = 1.61; $p = 0.20$	F = 0.11; $p = 0.73$	F = 0.01; $p = 0.91$
ANOVA - (Group x Muscle x Side)	F = 3.59; $p = 0.06$	F = 0.68; $p = 0.41$	F = 0.26; $p = 0.60$
Student t-test– (Intergroup comparison)	t = −0,44; $p = 0.65$	t = −0,75; $p = 0.45$	t = −2,13; $p < 0.05$*

*Significant difference; Inferential statistics for the jaw elevator muscles (masseter and temporalis) using mixed-model-ANOVA; Separate inferential statistics for the jaw depressors muscles (suprahyoid) using Student's t-test.

than on parafilm.

Thus, for future studies, it is suggested a standardization of the bite task for the comparison of the IEMG values of the masticatory muscles, i.e., MVC on parafilm.

The significant difference in non-normalized IEMG values between the right and left sides (without considering the groups or muscles) also merits attention. Such a finding may be explained by a preference for the right side over the left among the volunteers during the act of chewing. It is noteworthy that Pitta et al. (2015) defined that masticatory muscles are always bilaterally activated by the Central Nervous System and not unilaterally, since TMJ is a double-linked joint. In addition, previous studies showed that women with myogenous TMD have bilateral disorder of their joints (Berni et al., 2015; Rodrigues-Bigaton et al., 2017). Thus, although the bilateral EMG analysis of the present study is validated, the finding should be interpreted with caution, since no preference-side evaluation was performed during the act of chewing in the present study.

In this context, the findings of the present study have clinical significance for the health area, presenting potential to guide the treatment of women with myogenous TMD, since the disorder seems to reduce electrical activity in the masseter, but not in the anterior temporalis muscles. On the other hand, the disorder seems not to be related to the statistical difference found in general electrical activity on the muscular sides.

Some previous studies have evaluated EMG values of the suprahyoid muscles in the time domain. Berni et al. (2015) found significantly higher RMS values in the group with myogenous TMD than the control group during MVC on parafilm. In contrast, but confirming the findings of the present study, Liu et al. (1999) found no significant difference in non-normalized IEMG values of these muscles between volunteers with TMD and asymptomatic controls during MVC in intercuspal position. Therefore, it appears that the EMG values in the time domain of the suprahyoid muscle vary depending on how the EMG signal are processed, since the RMS parameter is calculated by the root mean square EMG signal, while the IEMG parameter is more dynamic, and represents the sum of the total area of the analyzed signal.

Thus, it is also suggested, that future studies evaluate MVC on parafilm and analyze non-normalized IEMG values of the masticatory muscles (jaw elevator and depressors) in order to standardize the analysis protocol in individuals with myogenous TMD.

Over the years, studies have characterized the EMG profiles of individuals with TMD. However, the present investigation offers novel information on IEMG values during MVC on parafilm and the first information on IEMG values for the suprahyoid muscles under this same condition. In addition, the electrical activity data from suprahyoid muscles are crucial to highlight how the antagonist muscles behave during MVC on parafilm, i.e., lower electrical activity compared to the agonist muscles (masseter and anterior temporalis). However, it is important to note that was observed absence of significant difference on the IEMG values of the suprahyoid muscles between myogenous TMD women and asymptomatic control.

Another important aspect to clarify, regards the reason for using normalized and non-normalized IEMG processing in this study. Many researchers recommend normalization, since non-normalized EMG data may confound the findings due to factors such as electrode positioning, different skin properties, muscle contraction velocity, muscle fiber type, body mass and size/shape of the muscle being evaluated (Al-Saleh et al., 2012). However, De Luca (1997) reports that the use of normalized EMG data tends to suppress distinctions between the data that would be associated with abnormal or pathological cases. Thus, the decision was made in the present study to evaluate EMG data with and without normalization, for which significant differences between the groups and muscles were only found with non-normalized data.

It is also worth noting that this research may help future studies that seek to evaluate EMG differences between individuals with myogenous TMD and asymptomatic control or even studies that seek evaluate the effects of treatment for myogenous TMD, suggesting the evaluation of the non-normalized EMG signal during MVC on parafilm. However, this research also suggests the analysis of the electrical activity of the masticatory muscles associated with the neuromuscular efficiency analysis, which is the ratio of the maximum bite force by the electrical activity value of each muscle, as was done by Zarrouk et al. (2012) on other muscles.

4.2. EMG analysis in frequency domain

In the frequency domain, significant difference between groups regarding MPF values was only found in the suprahyoid muscles, with higher values in the TMD group, a fact that brings new information on the firing rate of the motor units of the antagonist muscles of MVC on parafilm. This information may prove the mechanism of muscular disuse, that in the long term increases the concentration of fast fibers less resistant to fatigue in individuals with chronical pain of the masticatory muscles, as already established by Grünheid et al. (2009).

No previous studies were found in the literature on MPF values for the suprahyoids muscle during MVC on parafilm in individuals with myogenous TMD. However, Liu et al. (1999) analyzed MPF values for the right and left digastric muscles during MVC in intercuspal position and found no differences between individuals with TMD and asymptomatic controls. This difference can be due to the analysis of the MVC task in the intercuspal position by the authors above, since it could have altered the recruitment pattern of the motor units (Forrester et al., 2010), capturing units with

different firing rates.

In the present study, MPF values were significantly higher in the anterior temporalis muscles than the masseter muscles. Few studies have addressed frequency spectral characteristics in individuals with myogenous TMD during MVC without the aim of evaluating muscle fatigue. Tartaglia et al. (2011) found significantly lower MPF values for the masseter and temporalis muscles of individuals with arthrogenous TMD in comparison to asymptomatic controls during MVC for five seconds on cotton rolls, as well as significantly higher values for the temporalis muscles in comparison to the masseter muscles. These findings were partially confirmed in the present investigation, as higher MPF values were found for the temporalis muscles in comparison to the masseter muscles (without considering groups or sides). The differences found between the authors and the present investigation may be explained by the low pain intensity reported by the volunteers in the TMD group as well as the evaluation of myogenous rather than arthrogenous TMD, since individuals with arthrogenous TMD present changing the neuromuscular condition to allow some functions of the stomatognathic system (Epker et al., 1999).

Lauriti et al. (2013) found significant differences in median frequency values in young individuals with and without TMD during MVC on parafilm for 10 s, with higher values for the masseter and temporalis muscles of individuals with severe TMD in comparison to asymptomatic controls. According to the authors, the MPF values in individuals with severe TMD were increased due muscle hyperactivity, which was not found in the present study, possibly due to the low pain intensity and shorter analysis time (five seconds).

Finally, this study offers the combined characterization of IEMG, IEMGn and MPF values of the jaw elevators (masseter and temporalis muscles) and jaw depressors (suprahyoid muscles) during MVC on parafilm in women with myogenous TMD and asymptomatic controls. These data could be important for comparison purposes in future studies that address EMG profiles in individuals with arthrogenous TMD or those with myogenous TMD with greater pain intensity in the TMJ as well as after treatment for myogenous TMD, as widely divulged in the literature (Packer et al., 2015; Bortolazzo et al., 2015; Zhang et al., 2013; Gomes et al., 2014).

The present study has the following limitations: the evaluation of volunteers with low pain intensity in the myogenous TMD group, since a higher level of pain intensity could lead to even more pronounced changes in the EMG parameters of the masticatory muscles; the analysis of only volunteers with a diagnosis of myogenous TMD, since individuals with arthrogenous TMD could possibly exhibit greater pain and dysfunction, changing the neuromuscular condition to allow some functions of the stomatognathic system, which, in turn, would be evidenced in the EMG analysis; the lack of an evaluation of the preferential chewing side of the volunteers, validating the hypothesis that the higher difference in electrical activity on the right side of the jaw elevator muscles is really due to the masticatory preference of the volunteers; and the lack of the use of a bite force dynamometer for the evaluation of MVC, which could have revealed more signs of the disorder in the women of TMD group, allowing the analysis of the neuromuscular efficiency.

5. Conclusion

Based on the present findings, the hypothesis of the study was partially confirmed, as women with myogenous TMD and asymptomatic controls exhibited significant differences in non-normalized IEMG values for the masseter muscles and MPF values for the suprahyoid muscles during maximum voluntary clenching of the molars teeth on parafilm.

Conflicts of interest

The authors have no personal or financial conflicts of interest related to the present work.

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